

Parity

First of all I want to check the mass term with γ .

term 1 $\hat{M} = \begin{pmatrix} 0 & \gamma^\mu \\ \mu^T & 0 \end{pmatrix}$

$$-\frac{1}{2} \bar{\hat{q}} C \epsilon \hat{M} \hat{q}^T = \left\{ \hat{q} = \begin{pmatrix} q_R \\ \tilde{q}_R \end{pmatrix} = \begin{pmatrix} q_R \\ \epsilon C \bar{q}_L^T \end{pmatrix}, C = i\gamma^2\gamma^0 \right\}$$

$$= -\frac{1}{2} \begin{pmatrix} q_R \\ \epsilon C \bar{q}_L^T \end{pmatrix}^+ \gamma^0 C \epsilon \hat{M} \left[\begin{pmatrix} q_R \\ \epsilon C \bar{q}_L^T \end{pmatrix}^+ \gamma^0 \right]^T =$$

$$= -\frac{1}{2} (q_R^+, (\bar{q}_L^T)^+ C^+ \epsilon^+) \gamma^0 C \epsilon \hat{M} \underbrace{\gamma^0}_=\gamma^0 \begin{pmatrix} q_R^* \\ \epsilon^* C^* (\bar{q}_L^T)^* \end{pmatrix} =$$

$$= -\frac{1}{2} (q_R^+, ((q_L^+ \gamma^0)^T)^+ C^+ \epsilon^+) \gamma^0 C \epsilon \hat{M} \gamma^0 \begin{pmatrix} q_R^* \\ \epsilon^* C^* ((q_L^+ \gamma^0)^T)^* \end{pmatrix} =$$

$$= -\frac{1}{2} (q_R^+, (q_L^+ \gamma^0)^* C^+ \epsilon^+) \gamma^0 C \epsilon \hat{M} \gamma^0 \begin{pmatrix} q_R^* \\ \epsilon^* C^* [\gamma^0 (q_L^+)^T]^* \end{pmatrix}$$

$$= -\frac{1}{2} (q_R^+, (q_L^+ \gamma^0)^* C^+ \epsilon^+) \gamma^0 C \epsilon \hat{M} \gamma^0 \begin{pmatrix} q_R^* \\ \epsilon^* C^* \underbrace{\gamma^0}_{=\gamma^0} q_L^+ \end{pmatrix} =$$

$$= -\frac{1}{2} (q_R^+, q_L^T \gamma^0^* C^+ \epsilon^+) \gamma^0 C \epsilon \hat{M} \gamma^0 \begin{pmatrix} q_R^* \\ \epsilon^* C^* \gamma^0 q_L \end{pmatrix} =$$

$$= -\frac{1}{2} (q_R^+, q_L^T \gamma^0^* C^+ \epsilon^+) \underbrace{\gamma^0 C \gamma^0}_{=-C} \begin{pmatrix} 0 & \gamma^\mu \\ \mu^T & 0 \end{pmatrix} \begin{pmatrix} q_R^* \\ \epsilon^* C^* \gamma^0 q_L \end{pmatrix} =$$

$$\gamma^0 C \gamma^0 = i \underbrace{\gamma^0 \gamma^2 \gamma^0 \gamma^0}_{(\gamma^2)^+} = -i \gamma^2 \gamma^0 = -C$$

$$= \frac{1}{2} (q_R^+, q_L^T \gamma^0^* C^+ \epsilon^+) C \epsilon \begin{pmatrix} 0 & \gamma^\mu \\ \mu^T & 0 \end{pmatrix} \begin{pmatrix} q_R^* \\ \epsilon^* C^* \gamma^0 q_L \end{pmatrix} =$$

$$= \frac{1}{2} (q_R^+, q_L^T \gamma^0^* C^+ \epsilon^+) C \epsilon \begin{pmatrix} \gamma^\mu \epsilon^* C^* \gamma^0 q_L \\ \mu^T q_R^* \end{pmatrix}$$

$$= \underbrace{\frac{1}{2} q_R^\dagger C \epsilon \gamma \mu \epsilon^* C^* \gamma^0 q_L}_{(1)} + \underbrace{\frac{1}{2} q_L^T \gamma^0 C^+ \epsilon^+ C \epsilon \mu^T q_R^*}_{(2)}$$

$$(1) \quad \frac{1}{2} q_R^\dagger C \epsilon \gamma \mu \epsilon^* C^* \gamma^0 q_L = \frac{1}{2} q_R^\dagger \gamma^0 \gamma^0 C \epsilon \gamma \mu \epsilon^* C^* \gamma^0 q_L =$$

$$= \frac{1}{2} \bar{q}_R (\underbrace{\gamma^0 C C^* \gamma^0}_{=1}) \epsilon \gamma \mu \epsilon^* q_L = -\frac{1}{2} \bar{q}_R \underbrace{\epsilon \gamma \mu \epsilon^*}_{=1} q_L = -\frac{1}{2} \bar{q}_R \mu q_L \quad \checkmark$$

$$\gamma^0 : \gamma^2 \gamma^0 : \gamma^2 \gamma^0 \gamma^0 = -\gamma^0 \gamma^2 \gamma^0 \gamma^2 = \underbrace{\gamma^0 \gamma^0}_{1} \underbrace{\gamma^2 \gamma^2}_{-1} = -1$$

$$\left. \begin{array}{l} \epsilon^T = \gamma \epsilon \\ \epsilon^+ = \epsilon^{-1} \end{array} \right\} \Rightarrow \epsilon \epsilon^* = (\epsilon \epsilon^*)^T = \left[(\epsilon^*)^T \epsilon^T \right]^T = (\epsilon^+ \epsilon)^T = (\epsilon^{-1} \gamma \epsilon)^T = \gamma^T I = \gamma$$

$$\Rightarrow \gamma \epsilon \epsilon^* = \gamma^2 = 1$$

$$(2) \quad \frac{1}{2} q_L^T \gamma^0 C^+ \epsilon^+ C \epsilon \mu^T q_R^* = \frac{1}{2} q_L^T \gamma^0 C^+ \underbrace{\epsilon^+ C \epsilon}_{=1} \mu^T q_R^* =$$

$$= \frac{1}{2} q_L^T \gamma^0 \underbrace{C^+ C}_{=1} \mu^T q_R^* = \frac{1}{2} q_L^T \gamma^0 \mu^T q_R^* \quad \text{⊖}$$

$$C^+ C = (i \gamma_2 \gamma_0)^+ (i \gamma_2 \gamma_0) = \gamma_0^+ \gamma_2^+ \gamma_2 \gamma_0 = (\underbrace{\gamma_0 \gamma_0}_1) (\underbrace{\gamma_2 \gamma_2}_1) \gamma_2 \gamma_0 =$$

$$= \gamma_2 \gamma_0 \gamma_2 \gamma_0 = \underbrace{-\gamma_2 \gamma_2}_{-1} \underbrace{\gamma_0 \gamma_0}_1 = 1$$

$$\text{⊖} \quad \frac{1}{2} q_{L,d,i} \gamma_{\alpha\beta} (\mu^T)_{ik} q_{R\beta,k}^* = \{ \text{anticommute} \}$$

$\uparrow$ Lorents $\uparrow$ known

$$= -\frac{1}{2} q_{R\beta,k}^* \gamma_{\alpha\beta} (\mu^T)_{ik} q_{L,d,i} = -\frac{1}{2} q_{R\beta,k}^* (\gamma^0)^T_{\beta\alpha} (\mu)_{ki} q_{L,d,i} =$$

$$= -\frac{1}{2} q_R^\dagger \gamma^T \mu q_L = -\frac{1}{2} q_R^\dagger \gamma^0 \mu q_L = -\frac{1}{2} \bar{q}_R \mu q_L \quad \checkmark$$

→ give the 3rd term of (2.1)

Now we know that the 4th term in (2.5) must be h.c. of the 3rd one:

$$\begin{aligned} \left(-\frac{1}{2} \bar{q}_R \mathcal{M} q_L\right)^+ &= -\frac{1}{2} q_L^+ \mathcal{M}^+ (\bar{q}_R)^+ = -\frac{1}{2} q_L^+ \mathcal{M}^+ (q_R^+ \gamma^0)^+ = \\ &= -\frac{1}{2} q_L^+ \mathcal{M}^+ \gamma^0{}^+ q_R = -\frac{1}{2} q_L^+ \gamma^0 \mathcal{M}^+ q_R = -\frac{1}{2} \bar{q}_L \mathcal{M}^+ q_R. \end{aligned}$$

⇒ let's check if this happens for the 2nd term in (2.5):

$$\begin{aligned} \left(-\frac{1}{2} \hat{q}^T \hat{\mathcal{M}}^+ \varepsilon^* C \hat{q}\right)^+ &= -\frac{1}{2} \hat{q}^+ C^+ (\varepsilon^*)^+ \hat{\mathcal{M}} (\hat{q}^T)^+ = \\ &= -\frac{1}{2} \hat{q}^+ \gamma^0 C^+ \varepsilon^T \hat{\mathcal{M}} \gamma^0 (\hat{q}^T)^+ = \\ &= -\frac{1}{2} \bar{\hat{q}} (\gamma^0 C^+ \gamma^0) \varepsilon^T \hat{\mathcal{M}} \gamma^0 (\hat{q}^T)^+ = \left\{ \begin{array}{l} \varepsilon^T = \gamma \varepsilon \\ \Rightarrow \gamma \varepsilon^T = \gamma^2 \varepsilon = \varepsilon \end{array} \right\} \\ &= -\frac{1}{2} \bar{\hat{q}} (\gamma^0 C^+ \gamma^0) \varepsilon \hat{\mathcal{M}} (\hat{q}^+ \gamma^0)^T = \\ &= -\frac{1}{2} \bar{\hat{q}} \underbrace{(\gamma^0 C^+ \gamma^0)}_{\varepsilon} \hat{\mathcal{M}} \bar{\hat{q}}^T = -\frac{1}{2} \bar{\hat{q}} C \varepsilon \hat{\mathcal{M}} \bar{\hat{q}}^T \quad (\text{correct!}) \end{aligned}$$

$$\gamma^0 C^+ \gamma^0 = \gamma^0 (i \gamma^0)^+ \gamma^0 = \gamma^0 (-i) \gamma^0{}^+ \gamma^0 = -i \underbrace{\gamma^0 \gamma^0}_{1} \underbrace{(\gamma^0 \gamma^0)^+}_{1} \gamma^0 =$$

$$= -i \gamma^0 \gamma^2 = i \gamma^2 \gamma^0 = C$$

⇒ (4) is correct

$$\gamma \varepsilon^* = (\gamma \varepsilon)^* = (\varepsilon^T)^* = \varepsilon^+$$

$$(\varepsilon^T = \gamma \varepsilon)$$

Now we start to look @ UV the Lagrangian (2.5)

It has a symmetry, call it P: $q_R \leftrightarrow q_L$, $\mathcal{M} \leftrightarrow \mathcal{M}^\dagger$

$\Rightarrow$ This symmetry must exist at the level of (2.5):

$$\hat{q} = \begin{pmatrix} q_R \\ \tilde{q}_R \end{pmatrix} = \begin{pmatrix} q_R \\ \varepsilon C \bar{q}_L^T \end{pmatrix} \xrightarrow{P} \begin{pmatrix} q_L \\ \varepsilon C \bar{q}_R^T \end{pmatrix}.$$

How to realize P? Try:

$$\hat{q} \rightarrow \varepsilon C J \bar{\hat{q}}^T$$

$$\Rightarrow \hat{q} \rightarrow \varepsilon C J \begin{bmatrix} q_R \\ \varepsilon C \bar{q}_L^T \end{bmatrix}^T = \varepsilon \cdot C J \cdot \left[(q_R^\dagger \gamma^0, (\varepsilon C \bar{q}_L^T)^\dagger \gamma^0) \right]^T =$$

$$= \varepsilon C J \begin{pmatrix} \gamma^0 q_R^* \\ \gamma^0 \varepsilon^* C^* \bar{q}_L^{\dagger T} \end{pmatrix} = \varepsilon C J \begin{pmatrix} \gamma^0 q_R^* \\ \gamma^0 \varepsilon^* C^* \gamma_0 q_L \end{pmatrix} =$$

$$= \varepsilon C \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \gamma^0 q_R^* \\ \gamma^0 \varepsilon^* C^* \gamma_0 q_L \end{pmatrix} = \varepsilon C \begin{pmatrix} \gamma^0 \varepsilon^* C^* \gamma_0 q_L \\ \gamma^0 q_R^* \end{pmatrix} = C \begin{pmatrix} \gamma^0 C^* \gamma_0 q_L \\ \varepsilon \gamma^0 q_R^* \end{pmatrix} \quad \textcircled{=}$$

$$\varepsilon \varepsilon^* = \varepsilon (\varepsilon^\dagger)^T = (\varepsilon^T)^T (\varepsilon^\dagger)^T = (\varepsilon^\dagger \varepsilon)^T = (\varepsilon^\dagger \gamma_0 \varepsilon)^T = 1$$

$$C \gamma^0 C^* \gamma^0 = \underbrace{i \gamma^2 \gamma^0 \gamma^0 (-i)}_1 \underbrace{\gamma^{2*} \gamma^{0*} \gamma^0}_1 = \gamma^2 (-\gamma^2) = 1$$

$$\textcircled{=} \begin{pmatrix} q_L \\ \varepsilon C \gamma^0 q_R^* \end{pmatrix} = \begin{pmatrix} q_L \\ \varepsilon C \bar{q}_R^T \end{pmatrix}$$

correct ✓

$$\bar{q}_R^T = (q_R^\dagger \gamma^0)^T = \gamma^0 q_R^*$$

How should $\hat{\mathcal{M}}$ transform to preserve it?

$$\left[-\frac{1}{2} \hat{q}^T \hat{\mathcal{M}}^\dagger \varepsilon^\dagger C \hat{q} \right] \xrightarrow{P} -\frac{1}{2} (\varepsilon C J \bar{\hat{q}}^T)^T (\hat{\mathcal{M}}^\dagger)^\dagger \varepsilon^\dagger C (\varepsilon C J \bar{\hat{q}}^T) =$$

$$= -\frac{1}{2} \bar{q} J^T C^T \varepsilon^T (\hat{u}')^+ \underbrace{\varepsilon^+ C \varepsilon}_1 C J \bar{q}^T = -\frac{1}{2} \bar{q} J^T C \varepsilon^T (\hat{u}')^+ J \bar{q}^T \quad \textcircled{=}$$

$$C^T C C = i(\gamma_2 \gamma_0)^T i \gamma_2 \gamma_0 i \gamma_2 \gamma_0 = i^3 \gamma_0^T \gamma_2^T \underbrace{(-\gamma_2 \gamma_2)}_1 \underbrace{\gamma_0^2}_1 = -i \gamma_0 \gamma_2^T = -i \gamma_0 \gamma_2 = i \gamma_2 \gamma_0 = C$$

$$\left. \begin{array}{l} \gamma_0^+ = -\gamma_2 \\ \gamma_2^+ = -\gamma_0 \end{array} \right\} \Rightarrow \gamma_2^T = -\gamma_2^* = \gamma_2$$

$$\textcircled{=} -\frac{1}{2} \bar{q} J^T C^T \varepsilon^T (\hat{u}')^+ J \bar{q}^T \stackrel{!}{=} -\frac{1}{2} \bar{q} C \varepsilon \hat{u} \bar{q}^T$$

$$\Rightarrow J^T (\hat{u}')^+ J = \hat{u}$$

$$\Rightarrow (\hat{u}')^+ J = J \hat{u} \Rightarrow (\hat{u}')^+ = J \hat{u} J^T$$

$$\Rightarrow \hat{u}' = (J \hat{u} J^T)^+ = J \hat{u}^+ J^T$$

i.e.

$$\hat{u} \xrightarrow{P} J \hat{u}^+ J^T$$

(Same for $\psi = 2\text{Bo} \hat{u}$)

($u \xrightarrow{P} u^+$, Does this hold for our P ∇ without $\vec{x} \rightarrow -\vec{x}$)

$$J \quad \psi_+ = u + \psi J^T u^+ + u J \psi^+ \xrightarrow{P} u J \underbrace{\psi^+ J^T J^T}_1 u + u^+ J (J \psi^+ J^T)^+ u^+$$

$$= u J^2 \psi^+ u + u^+ \underbrace{J J}_1 \psi J^+ u^+ = u J \psi^+ u + u^+ \psi J^+ u^+ \Rightarrow P\text{-invariant} \quad \textcolor{orange}{!}$$

If we had minus this would be $\psi_- \rightarrow -\psi_-$

Now, we have to understand the symmetry transformation of ψ_μ .

$$\text{We have: } \bar{q} \cdot \gamma^\mu \psi_\mu \hat{q} \rightarrow (\varepsilon C J \bar{q}^T)^+ \gamma^0 \gamma^\mu \psi'_\mu \varepsilon C J \bar{q}^T =$$

$$\hat{q} \rightarrow \varepsilon C J \bar{q}^T$$

$$= (\bar{q}^T)^+ J^+ C^+ \underbrace{\varepsilon + \delta \delta^\mu V_\mu'}_{=J} \varepsilon C J \bar{q}^T = [(\hat{q} + \delta^0)^T]^+ J^+ C^+ \delta^0 \delta^\mu V_\mu' C J (\hat{q} + \delta^0)^T =$$

$$= \hat{q}^T \delta^0 J^+ \underbrace{C^+}_{=J} \underbrace{\delta^0 \delta^\mu V_\mu'}_{=J} C J \delta^0 \hat{q}^* =$$

$$\delta^0 C^+ \delta^0 = \delta^0 (i \delta^2 \delta^0)^+ \delta^0 = \underbrace{\delta^0 (-i)}_1 \delta^0 \delta^2 \delta^0 = -i (-\delta^2) \delta^0 = i \delta^2 \delta^0 = C$$

$$= \hat{q}^T C J \delta^\mu V_\mu' C J \delta^0 \hat{q}^* = \hat{q}^T \underbrace{(J^T \delta^{\mu T} V_\mu')_{\alpha\beta}}_{A_{\alpha\beta}^{ik}} \delta^0 \hat{q}^* \quad \odot \quad \hat{q}_{\alpha,i} A_{\alpha\beta}^{ik} \delta_{\beta\gamma}^0 q_{\delta,*}^* = \left\{ \begin{matrix} \text{ant.} \\ \text{comm.} \end{matrix} \right\}$$

$$C \delta^\mu C = i \delta_2 \delta_0 \delta^\mu \delta_2 \delta_0 = (\delta^\mu)^T$$

$$\left[\begin{array}{l} \mu=0 \Rightarrow -\delta_2 \delta_0^2 \delta_2 \delta_0 = -\delta_2^2 \delta_0 = \delta_0 = \delta_0^T \end{array} \right.$$

$$\mu=1 \Rightarrow -\delta_2 \delta_0 \delta^1 \delta_2 \delta_0 = -\delta_2 \delta^1 \delta_2 \delta_0^2 = -\delta_2 \delta^1 \delta_2 = \delta_2^2 \delta_1 = -\delta_1 = \delta_1^T$$

$$\mu=2 \Rightarrow -\delta_2 \delta_0 \underbrace{\delta_2 \delta_1 \delta_0}_{-1} = \delta_2 \delta_0 \delta_0 = \delta_2 = \delta_2^T$$

$$\mu=3 \xrightarrow{\text{same as } \mu=1} -\delta_3 = \delta_3^T$$

$$= -\hat{q}_{\delta,*}^* \delta_{\beta\gamma}^0 A_{\alpha\beta}^{ik} \hat{q}_{\alpha,i} = -\hat{q}_{\delta,*}^* \delta_{\beta\gamma}^0 (A^T)_{\beta\alpha}^{ki} \hat{q}_{\alpha,i} = -\hat{q}^+ \delta^0 A^T \hat{q} =$$

$$= -\bar{q} A^T \hat{q} = -\bar{q} (J^T \delta^{\mu T} V_\mu')^T \hat{q} = -\bar{q} J^T V_\mu'^T \delta^\mu J \hat{q} \stackrel{!}{=} \bar{q} \cdot \delta^\mu V_\mu \hat{q} \quad (\text{for inv})$$

$$\Rightarrow -J^T (V_\mu')^T \delta^\mu J = \delta^\mu V_\mu$$

$$\Rightarrow J^T (V_\mu')^T J = -V_\mu$$

$$V_\mu'^T = -J V_\mu J^T$$

$$\boxed{V_\mu' = -J V_\mu^T J^T} \quad (V_\mu \xrightarrow{P} V_\mu')$$

What happens with $V_{\mu 0}$?

$$\begin{aligned}
 V_{\mu 0} &= \partial_\mu V_0 - \partial_0 V_\mu - i[V_\mu, V_0] \stackrel{P}{\rightarrow} \partial_\mu (-J V_0^T J^T) - \partial_0 (-J V_\mu^T J^T) - \\
 &- i[-J V_\mu^T J^T, -J V_0^T J^T] = -J(\partial_\mu V_0^T - \partial_0 V_\mu^T) J^T - i[J V_\mu^T J^T, J V_0^T J^T] \\
 &= -J(\partial_\mu V_0^T - \partial_0 V_\mu^T) J^T - i(J V_\mu^T \cancel{J^T J} V_0^T J^T - J V_0^T \cancel{J^T J} V_\mu^T J^T) = \\
 &= -J(\partial_\mu V_0^T - \partial_0 V_\mu^T) J^T - i(J V_\mu^T V_0^T J^T - J V_0^T V_\mu^T J^T) = \\
 &= -J(\partial_\mu V_0^T - \partial_0 V_\mu^T) J^T - i J [V_\mu^T, V_0^T] J^T = \\
 &= -J \underbrace{(\partial_\mu V_0^T - \partial_0 V_\mu^T + i[V_\mu^T, V_0^T])}_{V_{\mu 0}^T} J^T = -J V_{\mu 0}^T J^T
 \end{aligned}$$

$$\left(\begin{aligned}
 V_{\mu 0} &= \partial_\mu V_0 - \partial_0 V_\mu - i[V_\mu, V_0] \\
 V_{\mu 0}^T &= \partial_\mu V_0^T - \partial_0 V_\mu^T - i[V_\mu^T, V_0^T] = \partial_\mu V_0^T - \partial_0 V_\mu^T - i[V_0^T, V_\mu^T] = \\
 &= \partial_\mu V_0^T - \partial_0 V_\mu^T + i[V_\mu^T, V_0^T]
 \end{aligned} \right)$$

ie. $V_{\mu 0} \rightarrow -J V_{\mu 0}^T J^T$

$$r_{\mu 0} = V_{\mu 0} \stackrel{P}{\rightarrow} -J r_{\mu 0}^T J^T = \ell_{\mu 0}$$

$$\ell_{\mu 0} = -J V_{\mu 0}^T J^T \stackrel{P}{\rightarrow} -J (-J V_{\mu 0}^T J^T)^T J^T = -J (-J V_{\mu 0} J^T) J^T = \underbrace{J^2}_{\mathbb{1}} V_{\mu 0} \underbrace{(J^T)^2}_{\mathbb{1}} = V_{\mu 0} = r_{\mu 0}$$

$\Rightarrow$

$$f_{\mu 0 \pm} = u \ell_{\mu 0} u^\dagger \pm u^\dagger r_{\mu 0} u \stackrel{P}{\rightarrow} u^\dagger r_{\mu 0} u \pm u \ell_{\mu 0} u^\dagger$$

$\Rightarrow f_{\mu 0, +}$ has a positive parity
 $f_{\mu 0, -}$ has negative parity

